



Calcium Phosphate Catalysts for Ethanol Coupling to Butanol and Butadiene

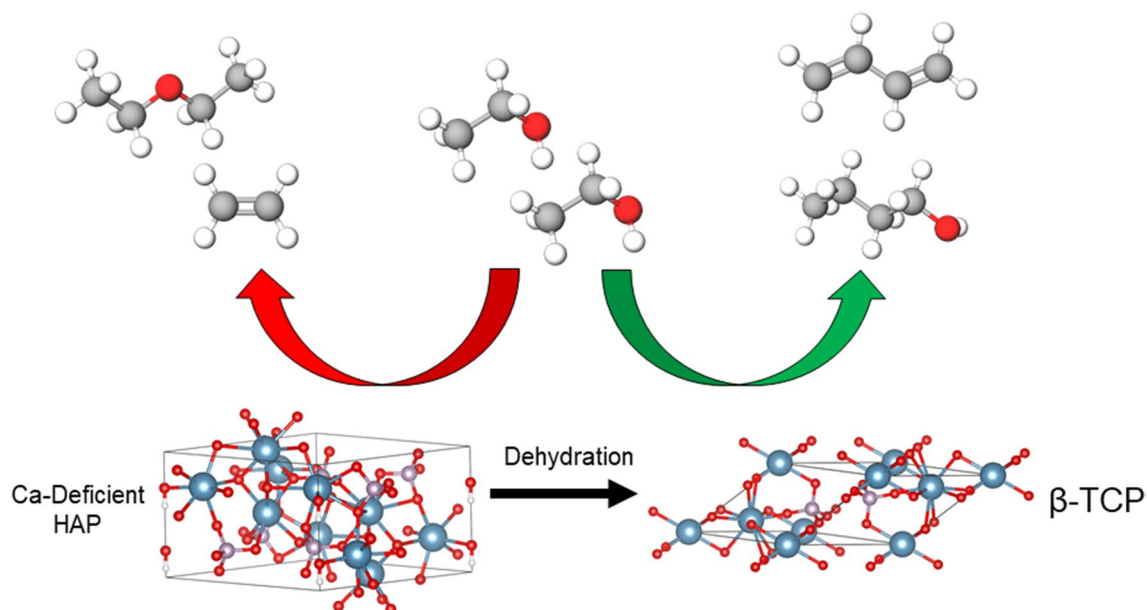
J. Tyler Prillaman¹ · Naomi Miyake¹ · Robert J. Davis¹

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Abstract

The catalytic conversion of ethanol to butanol and butadiene at 633 K and atmospheric pressure (7 vol% ethanol) was studied over calcium phosphate materials pretreated over a wide temperature range. For calcium phosphate pretreated at or below 923 K, the resulting solid had an apatite structure identified by X-ray diffraction and strong acid sites probed by triethylamine (TEA) adsorption microcalorimetry. The low temperature pretreated materials were not effective at ethanol coupling reactions but instead were highly selective to acid-catalyzed products ethene and diethyl ether. Pretreatment of calcium phosphate at 973 K transformed the apatite structure to β -tricalcium phosphate, which exposed both acid and base sites evaluated by TEA and CO₂ adsorption microcalorimetry. High temperature pretreated calcium phosphate catalysts formed the coupling products butanol and butadiene during ethanol reaction. The butadiene selectivity was improved by supporting calcium phosphate on zirconia.

Graphic abstract



The influence of catalyst structure on the activity and selectivity of calcium phosphates was elucidated for the conversion of ethanol.

Keywords Guerbet reaction · Lebedev reaction · Ethanol coupling · Adsorption microcalorimetry · Apatite

✉ Robert J. Davis
rjd4f@virginia.edu

Extended author information available on the last page of the article

1 Introduction

Butadiene (1,3-butadiene) is an important monomer for the polymer industry, particularly in the production of synthetic rubber and copolymers such as acrylonitrile–butadiene–styrene (ABS). Because the production of butadiene as a side-product of naphtha cracking has declined recently, alternative production routes of this monomer are being investigated. One approach is to convert ethanol to butadiene via the Lebedev process. This process is generally believed to occur through a cascade of reaction steps involving dehydrogenation of ethanol to acetaldehyde, self-condensation of acetaldehyde to crotonaldehyde, hydrogenation of crotonaldehyde to crotyl alcohol, and dehydration of crotyl alcohol to 1,3-butadiene [1–4].

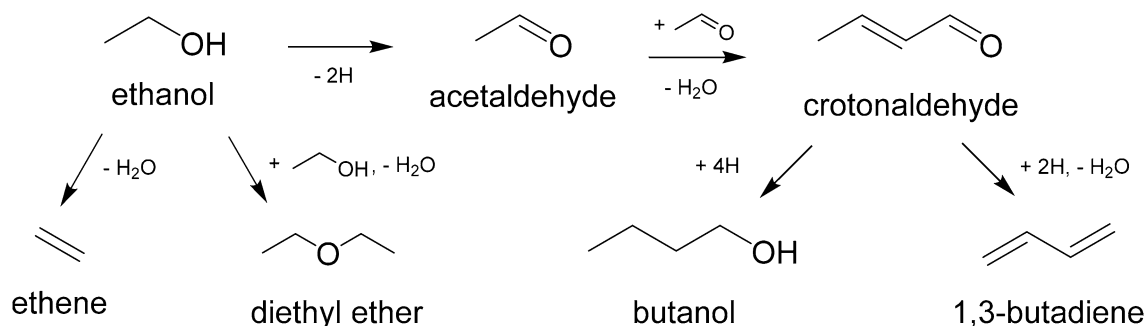
Many catalysts are reportedly capable of upgrading ethanol to butadiene, but side products such as ethene, diethyl ether, and butanol are often reported. Ethene and diethyl ether are typically formed through acid-catalyzed reactions of ethanol whereas butanol is likely formed in the Guerbet reaction, which proceeds through a similar cascade as the Lebedev reaction except the intermediate crotonaldehyde is completely hydrogenated to butanol [5, 6]. Scheme 1 illustrates the sequences that comprise the Lebedev and Guerbet reactions. The selectivity to butanol versus butadiene is likely dictated by the delicate balance between acidic and basic sites on the catalyst surface [1, 2, 7–9].

Mixed oxides and supported oxides have been investigated for the catalytic formation of butadiene from ethanol. The silica–magnesia (SiO_2 – MgO) system, which has relatively high selectivity to butadiene at modest conversion levels, is one of the most studied systems [8, 10–17]. Promotion of metal oxide catalysts with metals such as copper and silver increases the dehydrogenating capability of the catalyst, positively influencing the overall catalytic activity [16, 18]. Makshina et al. observed that promotion of SiO_2 – MgO with metals almost tripled the selectivity towards butadiene at conversions over 75% [15].

Another oxide of interest is ZrO_2 . Toussaint et al. reported that SiO_2 -supported ZrO_2 catalysts performed almost as well as the once industrially used tantalum-doped silica catalyst in the formation of butadiene from a mixture of ethanol and acetaldehyde [19]. Jones et al. attempted to tune the acidic and basic nature of the catalyst by exploring various combinations of mixed oxides. They found that the best performing catalyst for the Lebedev reaction was composed of a mixture of Zr, Zn, and SiO_2 [20]. A catalyst composed of Ag and ZrO_2 supported on SiO_2 has also been observed to exhibit high catalytic activity in the ethanol reaction, maintaining a selectivity around 70% at conversions greater than 80% [21, 22].

Hydroxyapatite (HAP; $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$) has been reported to be an effective catalyst for Guerbet coupling of ethanol to butanol, with a high selectivity to butanol of around 70% achieved at low conversion (5%) [23–26]. Hydroxyapatite has a tolerance for different cation-to-phosphorous ratios, allowing a metal-deficient synthesis. To balance the resulting charge due to cation deficiency, some of the hydroxyl groups are replaced by protons, which are typically represented in the stoichiometric formula as being associated with hydrogen phosphate ions (M is Ca, Sr, Ba): $M_{(10-x)}(\text{HPO}_4)_x(\text{PO}_4)_{(6-x)}\text{OH}_{(2-x)}$ [27–29]. The shift from the more basic hydroxyl to the more acidic hydrogen phosphate has been shown to shift selectivity from butanol to butadiene [5]. Furthermore, Osman et al. observed that washing and drying conditions used to prepare HAP affect its overall catalytic performance [30].

Other catalysts have also been studied, including titania, alumina, and zeolites, with and without promotion [9, 20, 21, 31–34]. Both SiO_2 – MgO and HAP serve as an example of catalysts exhibiting the balance between selectivity to butanol and butadiene. Specifically, MgO on its own tends to form butanol and less acid-catalyzed products, while addition of SiO_2 to MgO shifts selectivity significantly towards butadiene [1, 15, 34].



Scheme 1 Reaction network involved in Guerbet (butanol) and Lebedev (butadiene) reactions of ethanol

This paper investigates the effect of acidity on selectivity in the catalytic upgrading of ethanol to the C4 molecules butadiene and butanol. Since previous works have demonstrated how adding weakly acidic SiO_2 to basic MgO shifts selectivity from butanol to butadiene, we explored the potential changes in selectivity over hydroxyapatite by addition of ZrO_2 . For comparison MgO , SiO_2 , and various thermally-treated calcium phosphates and hydroxyapatites were also evaluated.

1.1 Results and Discussion

Table 1 summarizes the nominal composition and measured surface area of the catalysts prepared and evaluated in this study. The surface area of the calcium phosphate (CaP) catalysts decreased monotonically with increasing treatment temperature (see Table 1), from $71 \text{ m}^2\text{g}^{-1}$ for an untreated sample to $5.4 \text{ m}^2\text{g}^{-1}$ after thermal treatment at 1173 K. For these calcium phosphate samples, thermal treatment induced a phase change from an apatite structure, present at room temperature through 923 K, to mostly $\beta\text{-Ca}_3(\text{PO}_4)_2$ with remnants of apatite at 973 K and higher temperatures. Figure 1 presents the X-ray diffraction patterns of CaP before and after the various thermal treatments and clearly shows phase transition between 923 and 973 K.

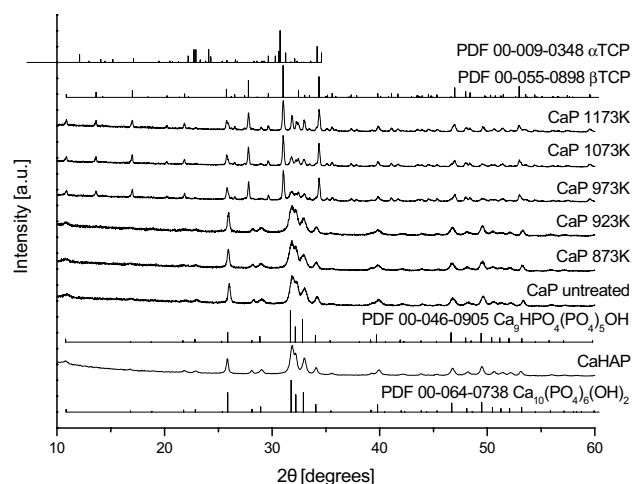


Fig. 1 X-ray diffraction patterns of calcium phosphate/apatite catalysts. Reference patterns (PDF) show peak location/intensity reported from their respective experimental patterns. Alpha-calcium phosphate is shown to indicate that the phases present in the tested catalysts are limited to apatite and beta only

The X-ray diffraction patterns in Fig. 1 indicate both CaP and CaHAP (treated at 873 K) have an apatite crystal structure even though their compositions should be different. The elemental compositions of the two samples were measured to verify the nominal compositions targeted in Table 1. The

Table 1 Summary of catalysts

Catalyst	Composition ^a	Support	Thermal treatment ^b (K)	Surface area ($\text{m}^2 \text{g}^{-1}$)
Reference materials				
MgO/SiO_2^c	1:1 Mg/Si	Syn- SiO_2^d	1073	17
CaHAP	$\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$	—	873	43
MgO	MgO	—	973	100
Syn- SiO_2^d	SiO_2	—	973	6
ZrO_2^e	ZrO_2	—	973	13
Calcium phosphate thermal treatments				
CaP untreated	$\text{Ca}_9(\text{HPO}_4)(\text{PO}_4)_5(\text{OH})$	—	none	71
CaP 873 K	$\text{Ca}_9(\text{HPO}_4)(\text{PO}_4)_5(\text{OH})$	—	873	54
CaP 923 K	$\text{Ca}_9(\text{HPO}_4)(\text{PO}_4)_5(\text{OH})$	—	923	43
CaP 973 K	$\text{Ca}_3(\text{PO}_4)_2$	—	973	20
CaP 1073 K	$\text{Ca}_3(\text{PO}_4)_2$	—	1073	8.7
CaP 1173 K	$\text{Ca}_3(\text{PO}_4)_2$	—	1173	5.4
Supported materials				
CaP/ SiO_2	$\text{Ca}_3(\text{PO}_4)_2$, 1:1 Ca/Si	Syn- SiO_2^d	973	12
CaP/ ZrO_2	$\text{Ca}_3(\text{PO}_4)_2$, 1:1 Ca/Zr	ZrO_2^e	973	17

^aCompositions assumed using nominal ratios from synthesis and phase identification by XRD

^b10 K min^{-1} ramp to temperature, held for 6 h under $100 \text{ cm}^3 \text{min}^{-1}$ flowing air

^cFormed by wet-kneading, $\text{Mg}(\text{OH})_2$ added to SiO_2 in water, stirred for 4 h

^dSilica formed by the hydrolysis of tetraethylorthosilicate (TEOS) at 308 K

^ePurchased from Aldrich, < 100 nm, monoclinic

nominal Ca/P ratio in CaHAP was 1.67, which is similar to that measured by elemental analysis (1.69) as reported in Table 2. The synthesis of CaP involved a nominal Ca/P ratio of 1.5 to eventually produce $\text{Ca}_3(\text{PO}_4)_2$, which is identical to the measured ratio listed in Table 2. Evidently, the chemical compositions of the phosphate samples were consistent with the desired compositions present in the synthesis procedure.

The influences of composition and thermal treatment on the catalytic performance of the calcium phosphate samples are presented in Tables 3 and 4. Calcium hydroxyapatite

(CaHAP) is well-recognized as an excellent catalyst for Guerbet coupling of ethanol to produce butanol and other C4 oxygenates [23, 24, 35, 36]. As expected, the reaction of ethanol over CaHAP under the experimental conditions used in this work (633 K, atmospheric pressure) was 77% selective to C4 oxygenates at 13% ethanol conversion, as shown in Table 3. The light hydrocarbons and diethyl ether, which are associated with acid sites on the catalyst, were produced in very low amounts (2% selectivity) over CaHAP, which

Table 2 Measured compositions of CaP materials by ICP-OES analysis

Catalyst	Sample (mg)	Calcium (wt%)	Phosphorus (wt%)	Zirconium (wt%)	Ca/P	Ca/Zr
CaHAP	52	39.8	18.2	—	1.69	—
CaP untreated	52	38.2	19.7	—	1.50	—
CaP/ZrO ₂	20	17.3	9.6	39.7	1.39	0.99

Table 3 Catalytic reaction of ethanol over CaHAP compared to reference materials

Catalyst	Temp (K)	Rate (mol m ⁻² s ⁻¹)	Conv. ^a (%)	Carbon selectivity (%)					
				Light HC/DEE ^b	Acetaldehyde	Butadiene	Butenes	C4 oxy ^c	Others ^d
CaHAP	633	6.1×10^{-8}	13	2	13	2	—	77	6
Syn-SiO ₂	633	4.2×10^{-9}	0.7 ^e	5	95	—	—	—	—
ZrO ₂	633	7.3×10^{-8}	14	58	33	2	—	4	3
MgO	633	2.0×10^{-9}	2	19	63	3	—	15	—
MgO/SiO ₂	633	8.1×10^{-9}	4	33	34	30	—	3	—

^aFlowrates selected to obtain low conversion values, calculated based on products

^bEthene, ethane, and diethyl ether

^cFour-carbon oxygenates: butanol, crotyl alcohol, butanal, crotonaldehyde

^dOthers: hexanol and/or unknowns (analyzed with an estimated molecular weight, carbon number, and response factor)

^eClose to background conversion in blank reactor

Table 4 Catalytic reaction of ethanol over CaP pretreated at various temperatures

Treatment ^a temperature	Temp (K)	Rate (mol m ⁻² s ⁻¹)	Conv. ^b (%)	Carbon selectivity (%)					
				Light HC/DEE ^c	Acetaldehyde	Butadiene	Butenes	C4 oxy ^d	Others ^e
None	633	1.4×10^{-8}	14	99	1	—	—	—	—
873 K	633	7.3×10^{-9}	15	97	3	—	—	—	—
923 K	633	6.3×10^{-9}	8	95	5	—	—	—	—
973 K	633	1.1×10^{-8}	13	21	8	9	1	60	1
1073 K	633	3.0×10^{-8}	10	11	9	3	1	75	1
1173 K	633	1.2×10^{-8}	4	14	23	4	—	59	—

^aThermally treated in flowing air for 6 h

^bFlowrates selected to obtain low conversion values, calculated based on products

^cEthene, ethane, and diethyl ether

^dFour-carbon oxygenates: butanol, crotyl alcohol, butanal, crotonaldehyde

^eOthers: hexanol and/or unknowns (analyzed with an estimated molecular weight, carbon number, and response factor)

is consistent with the previously reported weak acidity of CaHAP [37].

In contrast to ethanol conversion over CaHAP, the same reaction over CaP pretreated at or below 923 K produced mostly light hydrocarbons and diethyl ether with a selectivity of 95% or greater (see Table 4). These results confirm the presence of strong acid sites on the surface of the apatite structure of CaP that are not on CaHAP. Prior work with non-stoichiometric hydroxyapatite materials have identified Brønsted acid sites arising from the charge imbalance between Ca^{2+} and $(\text{PO}_4)^{3-}$ groups that require an extra proton to give $\text{Ca}_9(\text{HPO}_4)(\text{PO}_4)_5(\text{OH})$ [38–40]. Evidently, the presence of these Brønsted acid sites favors the unproductive dehydration of ethanol.

Thermal treatment of CaP at 973 K and higher temperatures dehydrated the material and transformed it to a β -tricalcium phosphate ($\text{Ca}_3(\text{PO}_4)_2$) structure. The high temperature treatment effectively decreased the acidic nature of the lower temperature treated samples, thus shifting the product selectivity during ethanol conversion. Table 4 presents the selectivity of ethanol reactions over CaP treated at 973 K, 1073 K, and 1173 K. The selectivity to light hydrocarbons and diethyl ether decreased from 21 to 11% (at 13 and 10% conversion, respectively) by simply increasing the thermal treatment temperature from 973 to 1073 K.

The reactivity results clearly show a significant change in the acid–base nature of the CaP catalysts between 923 and 973 K, the exact interval over which the fresh catalyst $\text{Ca}_9(\text{HPO}_4)(\text{PO}_4)_5(\text{OH})$ dehydrated and transformed to $\text{Ca}_3(\text{PO}_4)_2$. In an attempt to correlate the performance of the catalysts with the acid and base sites generated by thermal treatment, adsorption microcalorimetry of triethylamine (TEA) and carbon dioxide was used to probe the acid and base sites, respectively. Figure 2 shows the adsorption isotherms and the difference in the heats of adsorption of TEA over the CaP catalyst treated at 923 K and 973 K. Whereas the saturation uptake of TEA ($2 \mu\text{mol m}^{-2}$) on CaP was unchanged by the structural transformation between 923 and 973 K, the sample treated at 973 K exhibited heats of TEA adsorption about 30 kJ mol^{-1} lower than those for the sample treated at 923 K (Fig. 2). The greater TEA adsorption energy on $\text{Ca}_9(\text{HPO}_4)(\text{PO}_4)_5(\text{OH})$ relative to that on $\text{Ca}_3(\text{PO}_4)_2$ is consistent with the very high selectivity (95%) to acid-catalyzed products during ethanol conversion over the former sample (Table 4). The heats of CO_2 adsorption on CaP pretreated at 923 K or lower were too low to reliably measure suggesting very low basic strengths on these materials, which is consistent with the lack of coupling products observed over $\text{Ca}_9(\text{HPO}_4)(\text{PO}_4)_5(\text{OH})$ as reported in Table 4. Pretreatment of CaP at 973 K to form $\text{Ca}_3(\text{PO}_4)_2$ resulted in a catalyst that adsorbed CO_2 as reported in Fig. 3, with a saturation coverage of about $0.7 \mu\text{mol m}^{-2}$ and heats of adsorption ranging from about $100\text{--}125 \text{ kJ mol}^{-1}$. The major

products over these high temperature treated catalysts are C4 oxygenates from the Guerbet coupling of ethanol. Interestingly, the acid character of CaP that remains after treatment at 973 K is adequate to produce butadiene (and butenes) via the Lebedev reaction, with a combined selectivity of 10% at a conversion of 13% ethanol. Treatment of CaP at 1073 K reduced the selectivity to all of the products associated with acid sites (light hydrocarbons, diethyl ether, butadiene, and butenes) in favor of C4 oxygenates. Clearly, the thermal treatment can be used to adjust the acid–base nature of the calcium phosphate catalysts.

To further explore the balance of surface and acid base sites necessary for producing butadiene from ethanol, CaP was mixed with SiO_2 (a weakly acidic oxide) and with ZrO_2 (an amphoteric oxide). Table 3 presents the catalytic performance of the individual SiO_2 and ZrO_2 catalysts in ethanol conversion, illustrating the very poor activity of silica (very low areal rate, near background conversion) and the high activity of zirconia for acid-catalyzed dehydration reactions. Given the amphoteric nature of ZrO_2 , it is not surprising that small amounts of C4 oxygenates and C4 olefins are produced from the ethanol reaction, but the selectivities to major products are 58% to light hydrocarbons and diethyl ether, and 33% to acetaldehyde, at an ethanol conversion of 14%. Table 5 summarizes the catalytic performance of CaP/ SiO_2 and CaP/ ZrO_2 in the conversion of ethanol at 633 K. Addition of SiO_2 to CaP had little influence on the selectivity to products as the composite material produced mostly C4 oxygenates. In contrast, addition of ZrO_2 to CaP created a composite catalyst with suitable acid and base sites to produce butadiene with 19% selectivity at 10% ethanol conversion. Doubling the conversion to 22% also produced butadiene with 20% selectivity. The C4 oxygenate selectivity was low ($<5\%$) at both levels of conversion suggesting that C4 coupling intermediates such as crotonaldehyde (see Scheme 1) that were formed on the composite catalyst eventually transformed to butadiene. The acidity of the amphoteric ZrO_2 component of the mixed catalyst was likely responsible for the high selectivity (57–63%) to light hydrocarbons and diethyl ether. These results suggest that it may be possible to optimize the ratio of ZrO_2 and CaP in the composite material to improve butadiene selectivity.

The performance of the composite CaP/ ZrO_2 catalyst can be compared with MgO/SiO_2 , a previously well-studied material for the Lebedev reaction of ethanol to butadiene [8, 10–17]. Table 3 shows the catalytic performance of the separate MgO and SiO_2 components, illustrating that neither oxide is effective for producing butadiene or C4 oxygenates. The composite catalyst MgO/SiO_2 produced butadiene with 30% selectivity at only 4% conversion of ethanol, which is consistent with prior reports on the efficiency of this catalyst for the production of butadiene [15]. Whereas the areal rate of ethanol conversion was five times greater over CaP/

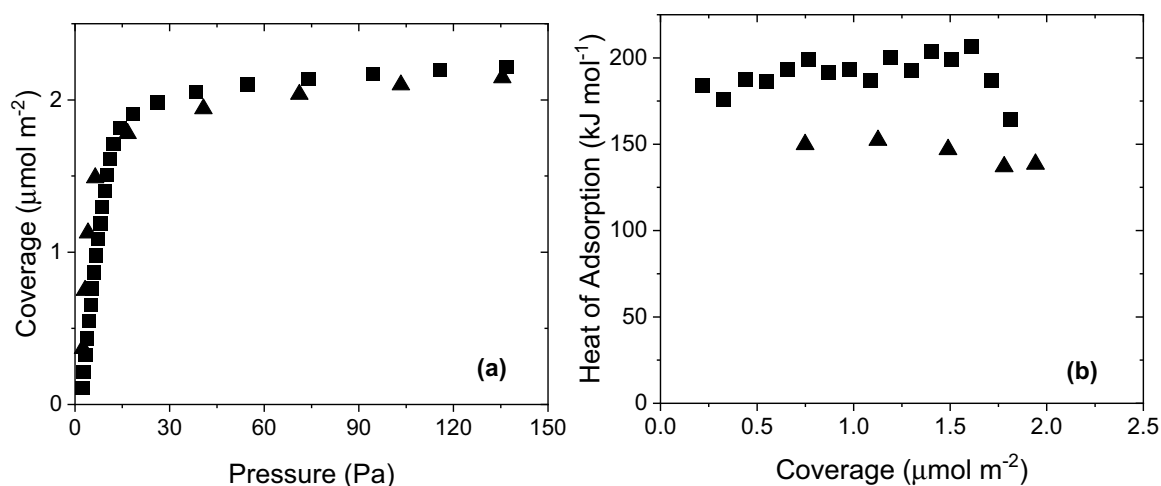


Fig. 2 Adsorption microcalorimetry of TEA at 308 K on CaP pretreated at 923 K (■) and 973 K (▲). **a** Adsorption isotherm. **b** Differential heat of adsorption as a function of coverage

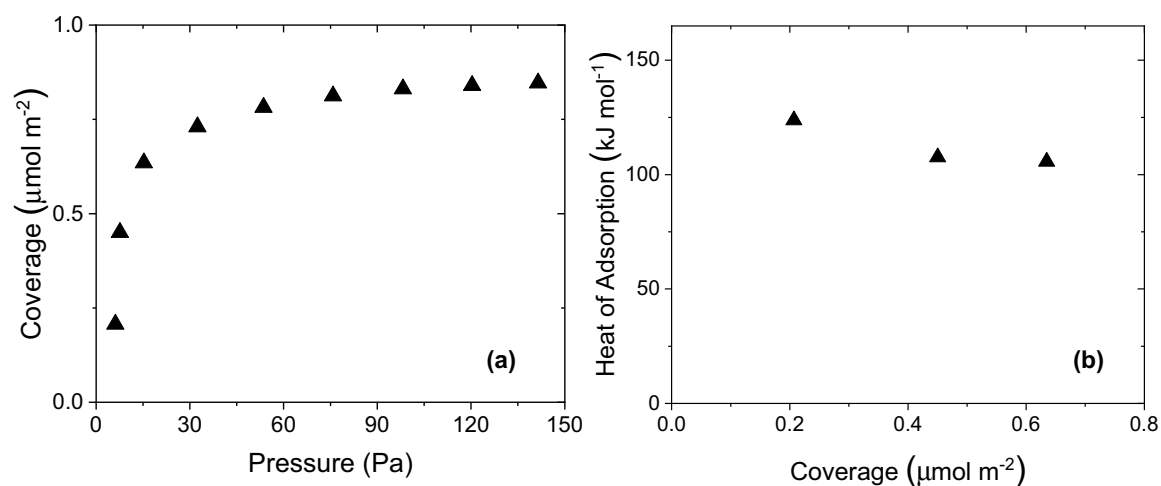


Fig. 3 Adsorption microcalorimetry of CO₂ at 308 K on CaP pretreated at 973 K (▲). **a** Adsorption isotherm. **b** Differential heat of adsorption as a function of coverage.

Table 5 Catalytic reaction of ethanol over CaP mixed with various metal oxides treated at 973 K. Flowrates and catalyst loading were varied to obtain low/high conversion values

Catalyst	Temp. (K)	Rate (mol m ⁻² s ⁻¹)	Conv. (%)	Carbon selectivity (%)					
				Light HC/ DEE ^a	Acetaldehyde	Butadiene	Butenes	C4 oxy ^b	Others ^c
CaP/SiO ₂	633	3.4 × 10 ⁻⁸	6	9	31	6	—	54	—
CaP/ZrO ₂	633	4.1 × 10 ⁻⁸	10	57	17	19	2	4	1
CaP/ZrO ₂	633	3.0 × 10 ⁻⁸	22	63	9	20	3	4	1

^aEthene, ethane, and diethyl ether.

^bFour-carbon oxygenates: butanol, crotyl alcohol, butanal, crotonaldehyde.

^cOthers: hexanol and/or unknowns (analyzed with an estimated molecular weight, carbon number, and response factor)

ZrO₂ compared to MgO/SiO₂, the selectivity to butadiene was greater over MgO/SiO₂.

2 Conclusions

Synthetic calcium hydroxyapatite Ca₁₀(PO₄)₆(OH)₂ thermally treated at 873 K has a surface area of 43 m² g⁻¹ and catalyzed the reaction of ethanol to C4 oxygenates with 77% selectivity at 633 K. A calcium-deficient hydroxyapatite Ca₉(HPO₄)(PO₄)₅(OH) converted to β-Ca₃(PO₄)₂ during thermal treatment between 923 and 973 K, which reduced the surface area from 43 to 20 m² g⁻¹. The acidic nature of calcium-deficient hydroxyapatite was confirmed by the heat of adsorption of triethylamine (~180–200 kJ mol⁻¹), very low heat of CO₂ adsorption (not quantifiable), and very high selectivity (≥95%) of ethanol reaction to the acid products ethene and diethyl ether. Thermal transformation to β-Ca₃(PO₄)₂ at 973 K produced a catalyst with both acidic and basic sites as evaluated by triethylamine and CO₂ adsorption microcalorimetry, and the reaction of ethanol favored the production of C4 oxygenates (60%) with a small amount of butadiene (9%). Whereas the areal uptake of triethylamine was unaffected by the thermal transformation, the heat of adsorption was about 30 kJ mol⁻¹ lower on β-Ca₃(PO₄)₂ compared to Ca₉(HPO₄)(PO₄)₅(OH), which is consistent with the relatively low selectivity to acid-catalyzed products ethene and diethyl ether (≤21%) in the catalytic reactions of ethanol over β-Ca₃(PO₄)₂. Improvement in the butadiene selectivity to 20% was achieved by addition of amphoteric ZrO₂ to a calcium phosphate sample.

3 Experimental Section

3.1 Catalyst Synthesis

Calcium hydroxyapatite (CaHAP) was prepared according to the method described by Tsuchida et al. to form stoichiometric apatite, with a Ca/P ratio of 1.67 [5, 41]. In summary, ammonium phosphate solution (0.3 M, using (NH₄)₂HPO₄, Aldrich ≥99.99% trace metals basis) and calcium nitrate solution (0.5 M, using Ca(NO₃)₂·4H₂O, Acros Organics ACS reagent) were both added dropwise to distilled, deionized (DDI) water at 333 K to initiate precipitation. The pH of the solution was maintained at 10 by dropwise addition of ammonium hydroxide (Fisher 14.8 N certified ACS plus). Both precursor solutions were adjusted to pH of 10 before addition to water. The precipitate was recovered by filtration, rinsed with DDI water to remove excess ions, and dried overnight in air at ~390 K. The sample was then heated at 10 K min⁻¹ to 873 K in flowing air (100 cm³ min⁻¹) and

held at that temperature for 2 h. The sample was then sieved to 106–180 μm particle sizes prior to loading in the reactor.

A series of calcium phosphate Ca₃(PO₄)₂ (CaP) catalysts was prepared from one batch of precursor material that was treated at different temperatures. The precursor material, denoted here as CaP untreated, was formed by adding calcium nitrate (Acros) solution (0.3 M) dropwise to 0.2 M ammonium phosphate (Aldrich) at room temperature using ammonium hydroxide (Fisher) added dropwise to maintain a pH of 9. Both solutions were adjusted to a pH of 9 before the precipitation. The precipitate was recovered by filtration, rinsed with DDI water, and dried overnight in air at 390 K. Thermal treatment of CaP untreated formed the catalysts denoted as CaP 873 K, CaP 923 K, CaP 973 K, CaP 1073 K, and CaP 1173 K, where the treatment temperature is listed in the labels. Each thermal treatment was performed in flowing air (100 cm³ min⁻¹) using a 10 K min⁻¹ ramp followed by constant treatment temperature for 6 h. Samples were sized to 106–180 μm prior to loading into the reactor.

The untreated calcium phosphate (CaP untreated) was also mixed with synthetic SiO₂ or monoclinic ZrO₂ (Aldrich, <100 nm) in an attempt to add some acid functionality to the composite catalyst. Silica was prepared by the modified Stöber route as reported by Angelici et al. [16] Tetraethylorthosilicate (TEOS, Aldrich 98% reagent) was added to a solution of 5:1 ethanol (Sigma-Aldrich ≥99.5% ACS reagent) and aqueous ammonia (Fisher) to give a final solution that was 0.34 M in TEOS. The TEOS/ethanol/ammonia solution was heated to 353 K in a sealed vessel and held at that temperature overnight with stirring. The resulting solid was recovered by solvent removal via rotary-evaporation until dry. Ethanol was then used to remove the powder from the reaction vessel, and the solid was dried overnight in an oven. This silica sample is denoted as Syn-SiO₂. The mass of oxide (either SiO₂ or ZrO₂) used relative to CaP untreated was selected to give a 1:1 molar ratio of calcium to silicon/zirconium in the final mixed catalyst. Mixing of solids occurred in just enough DDI water to suspend both the CaP and support; typically 10 cm³ of water was used for a batch of around 1 g of product. This slurry was stirred for 4 h before being dried in air at ~390 K. The dried catalyst was then thermally treated as described above, holding at 973 K for 6 h, and then sized to 106–180 μm. The mixed catalyst is denoted as CaP/SiO₂ or CaP/ZrO₂ respectively.

Before reaction, the monoclinic ZrO₂ was thermally treated as described above at 973 K for 6 h and sized to 106–180 μm.

A mixture of magnesia and silica (MgO/SiO₂) was synthesized by combining Mg(OH)₂ with Syn-SiO₂ using the same method described for CaP/SiO₂, which is similar to that used by Angelici et al. [16] The Mg(OH)₂ was

first prepared by adding a 3:1 excess of 1.0 M ammonium hydroxide (Fisher) dropwise to 0.5 M magnesium nitrate ($\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, Acros Organics 98 + %). The resulting slurry was stirred overnight and the $\text{Mg}(\text{OH})_2$ precipitate was filtered, rinsed with DDI water, and dried in air at ~ 390 K overnight. The $\text{Mg}(\text{OH})_2$ was then added to Syn-SiO_2 in a 1:1 molar ratio of Mg to Si in DDI water as described above. The recovered mixture of $\text{Mg}(\text{OH})_2/\text{SiO}_2$ was dried and then thermally treated in flowing air at 1073 K for 6 h to convert the hydroxide to oxide. The MgO/SiO_2 sample was sized to 106–180 μm before loading into the reactor.

Magnesia (MgO) was prepared from $\text{Mg}(\text{OH})_2$ precursor that was prepared according to the methods described above. Thermal treatment at 973 K for 6 h decomposed the $\text{Mg}(\text{OH})_2$ to MgO .

3.2 Catalyst Characterization

Elemental composition of the calcium phosphate materials was determined by Galbraith Labs (Knoxville, TN, USA) using ICP-OES analysis. The surface areas of the catalysts were estimated by the BET method using N_2 adsorption isotherms collected on a Micromeritics ASAP 2020 adsorption system. The X-ray diffraction patterns of the catalysts were recorded on a Panalytical Empyrean X-Ray Diffractometer (XRD) using $\text{Cu K}\alpha$ radiation and a scan rate of $0.0895^\circ \text{ s}^{-1}$ over a 2θ range of 10 – 60° .

Acid and base sites were characterized by studying the adsorption of triethylamine (TEA) and carbon dioxide (CO_2) on the calcium phosphate materials using a heat-flow microcalorimeter. Acid sites were probed using the base TEA while basic sites were probed using CO_2 , a weak acid. The samples were outgassed at 773 K for 4 h under vacuum before the samples were dosed with either TEA or CO_2 . All measurements were taken at 308 K.

3.3 Catalytic Reaction of Ethanol

Reactions were performed with a packed bed titanium reactor in a down-flow configuration. The observed background conversion of ethanol (Koptec anhydrous) was less than 1% at all temperatures tested. Ethanol was introduced to the system by syringe pump to create a reactant stream of 7% ethanol by volume in N_2 . The total flow rate and mass loading of catalyst were used to adjust conversion levels.

Prior to reaction, the catalyst was pretreated in situ in $100 \text{ cm}^3 \text{ min}^{-1}$ flowing N_2 at 773 K for 1 h. After this pretreatment, the reactor was cooled under N_2 flow to the reaction temperature of 633 K. A Shimadzu GC-2014 equipped with a flame-ionization detector (FID) was used to provide quantitative information on the products formed. A flow of

$5 \text{ cm}^3 \text{ min}^{-1}$ methane into the product stream was used as an internal standard for GC analysis. Although not all products were identified, the estimated selectivity to unknown components was low ($< 3\%$) for all tests. The unknowns were assigned an estimated molecular weight and carbon number based on retention time to make this assessment.

Conversions and selectivities are based on the amount of carbon present in the product. Specifically, conversion is calculated by determining the molar amount of carbon in the products and dividing by the amount of carbon fed per unit time:

$$X = \sum_{i=1}^k \left(\frac{c_i n_i}{2n_{\text{EtOH}}} \right) \times 100$$

where X is conversion, c_i is the number of carbons in product i , n_i is the number of moles of product i formed, n_{EtOH} is the initial number of moles of ethanol, and k is the number of products detected. Selectivity is defined as the amount of product carbon in a specific component divided by the total amount of carbon in all of the products detected:

$$S_i = \left(\frac{c_i n_i}{\sum_{i=1}^k c_i n_i} \right) \times 100$$

where S_i is the selectivity of component i . Values presented here were measured at the steady state.

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Compliance with ethical standards

Conflict of Interest All authors declare that they have no conflict of interest.

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Affiliations

J. Tyler Prillaman¹ · Naomi Miyake¹ · Robert J. Davis¹ 

¹ Department of Chemical Engineering, University of Virginia, 102 Engineer's Way, Charlottesville, VA 22904-4741, USA